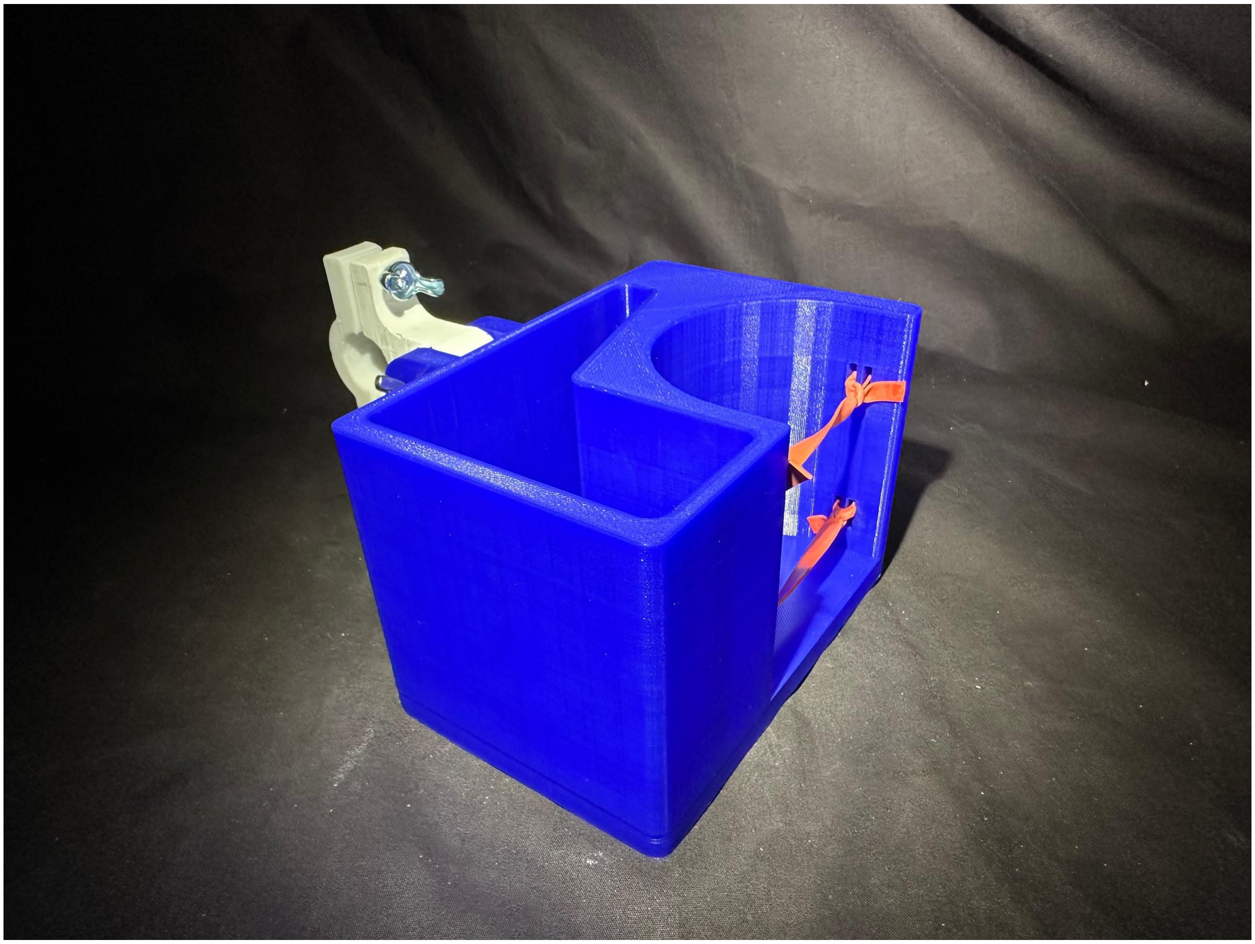
UC Berkeley

ENGIN-29 Final Project Report



Luke Fewx, Yankang Liu, Drew McCall, Armando Morales, Jason Wang

ENGIN-29 Manufacturing and Design Communication

Sara McMains

7 Dec. 2023

B. Comprehensive Description of Contributions

Lab 7

Everyone brainstormed different ideas to base the project around. Luke brought up the idea of focusing our project on being able to address storage on human transportation. Everyone brainstormed ideas to solve this problem. We ultimately ended up deciding on creating a compartment that could be attached to a bike or scooter to solve this problem. Armando worked on the lab report for the week and it was read through, edited, and approved by the others.

Lab 8

Jason, Drew, and Armando worked on the collaborative plan prompts for the Teaming Document because Yankang and Luke were late to the lab. We then got our object for analyzing, which was a hand-held hole puncher, and a standard desk hole puncher. Everyone helped in creating the reverse-engineering tables for both objects, and the bill of materials. Armando set the Lab Report up, read through, edited, and approved by the rest of the team.

Lab 9

We finalized our project idea after discussion on whether or not we should start from scratch. Yankang drew the detailed sketch of the product on his iPad. Yankang, Luke, Drew, and Jason worked on the Fits table for drawings Yankang had produced. Drew and Armando worked on the Lab Report this week.

Lab 10

For this Lab, the Lab report was split between Drew and Yankang. Drew recorded data in spreadsheets for other group members, used plastic calipers and micrometers to measure dimensions. He also worked on questions 2, 7, and 8 in the lab report. Jason and Yankang participated in drop gauge, micrometer, plastic caliper, and digital caliper measurements.

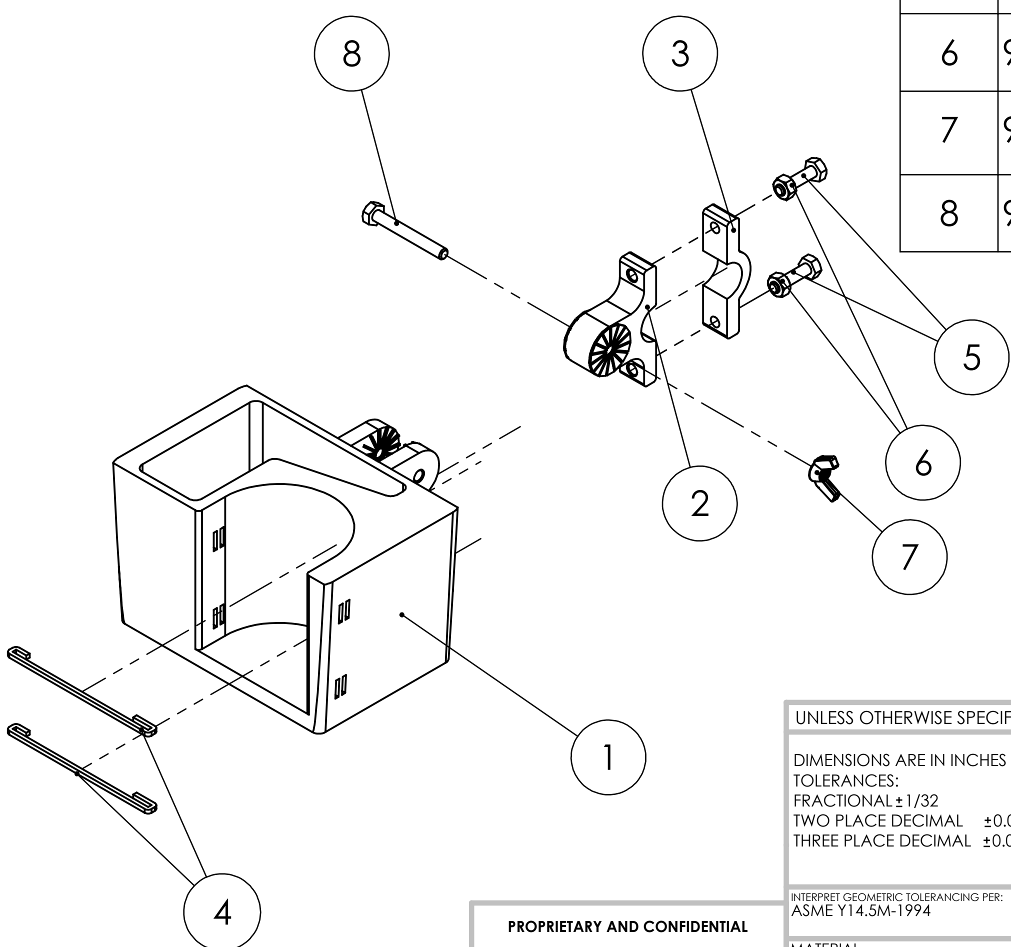
Lab 11

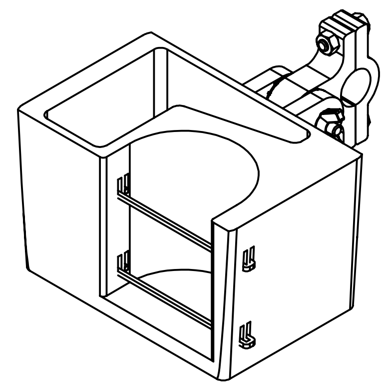
In this lab, Yankang revised his detailed drawing and we discussed team roles moving forward. Everyone contributed to working on the components and candidate materials and processes. Drew completed the fits and tolerances table. Yankang worked on the SolidWorks CAD models for all components and their respective drawings. We assigned roles to each team member.

Tasks since Lab 11

Since lab 11, a lot has been done. Yankang created the CAD files for our three components. Yankang also made the exploded isometric assembly drawing, the outline assembly drawing, and the bill of materials. Drew made the detail drawings and contributed to the GD&T. Jason added some extra details to the detail drawings. Drew purchased the hardware and Armando made an additional hardware run. Jason, Drew, and Yankang printed the components. Drew updated and edited the lab report. The entire group showed up to the Jacobs showcase.

ITEM NO.	PART NUMBER	DESCRIPTION	MATERIAL	QTY.
1	E29-G1028-1	Main Tray	Nylon	1
2	E29-G1028-2	Clamp 1	Nylon	1
3	E29-G1028-3	Clamp 2	Nylon	1
4	E29-G1028-4	Rubber Band	Rubber	2
5	92240A542	HBOLT 0.2500-20x1x0.75-N	Stainless Steel	2
6	91845A029	HNUT 0.2500-20-D-N	Stainless Steel	2
7	92001A321	18-8 Wing Nut	Stainless Steel	1
8	92240A550	HBOLT 0.2500-20x2x0.75-N	Stainless Steel	1





UNLESS OTHERWISE SPECIFIED:		NAME	DATE
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	CHECKED	DM	11/30
	ENG APPR.		
	MFG APPR.		
INTERPRET GEOMETRIC TOLERANCING PER: ASME Y14.5M-1994	Q.A.		
MATERIAL See BoM	COMMENTS: THIRD ANGLE PROJECTION		
DO NOT SCALE DRAWING			

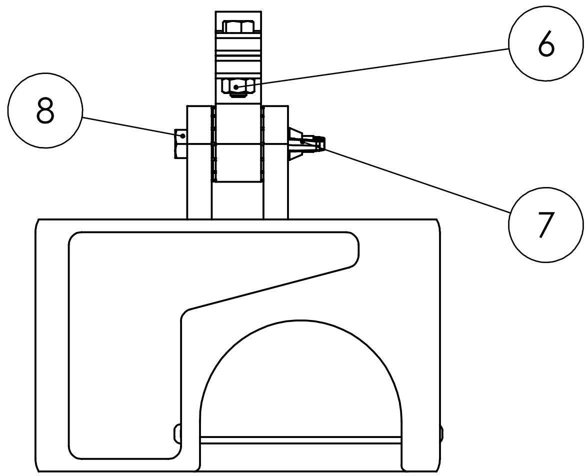
Great Tolerance Factory Organization
E29-G1028

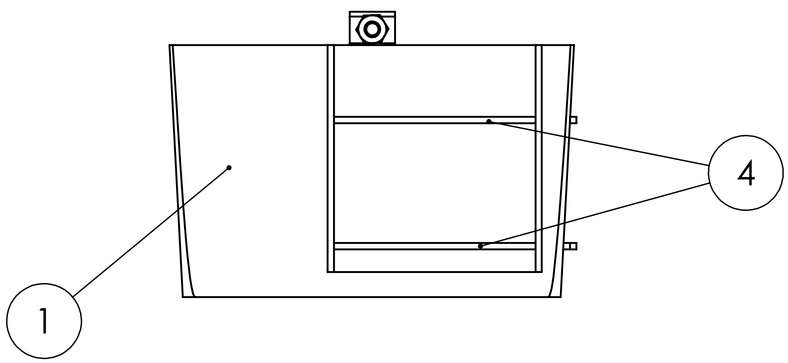
TITLE:

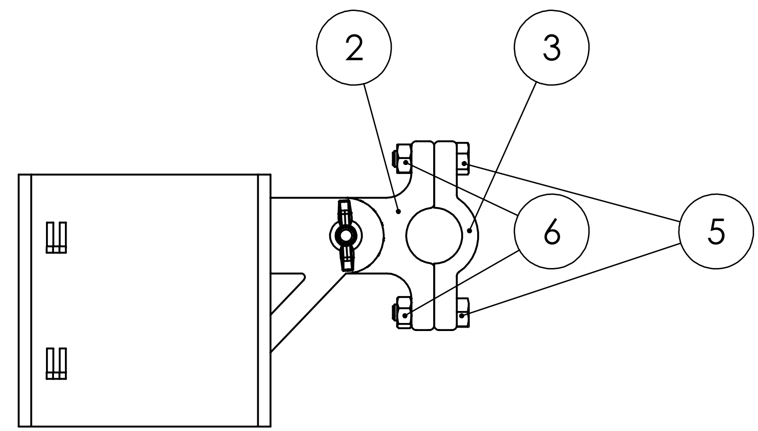
Assembly Exploded View with Bill of Materials

SIZE	PART NO.	REV
A	E29-G1028-000-1	1

SCALE: 1:4 | SHEET 1 OF 2







UNLESS OTHERWISE SPECIFIED:		NAME	DATE	Great Tolerance Factory Organization E29-G1028
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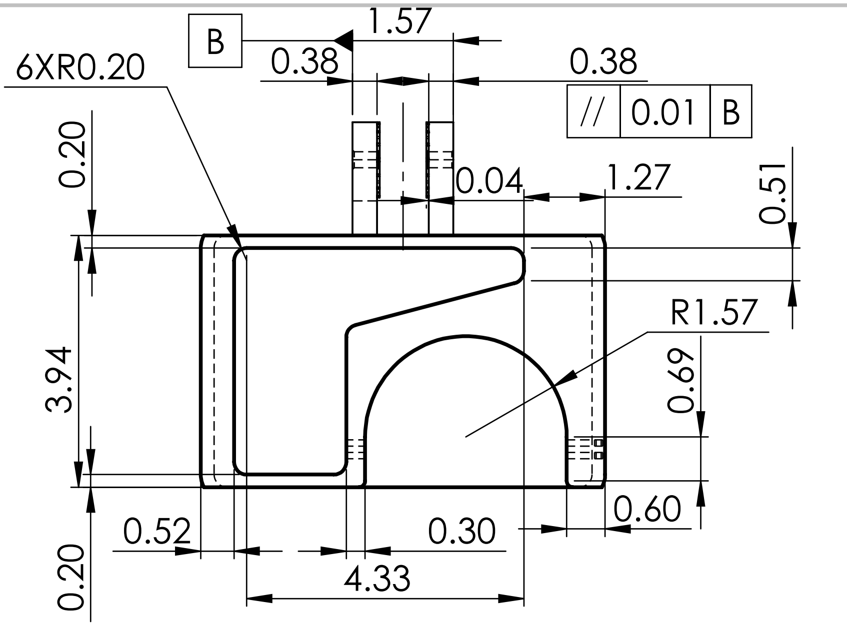
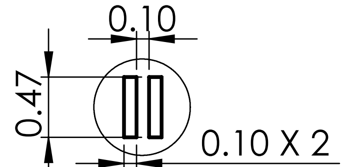
	CHECKED	DM	11/30	
	ENG APPR.			
	MFG APPR.			
INTERPRET GEOMETRIC TOLERANCING PER: ASME Y14.5M-1994	Q.A.			
MATERIAL See BoM	COMMENTS: THIRD ANGLE PROJECTION			SIZE A · PART NO. E29-G1028-000-2 · REV 1
DO NOT SCALE DRAWING				SCALE: 1:3 · SHEET 2 OF 2

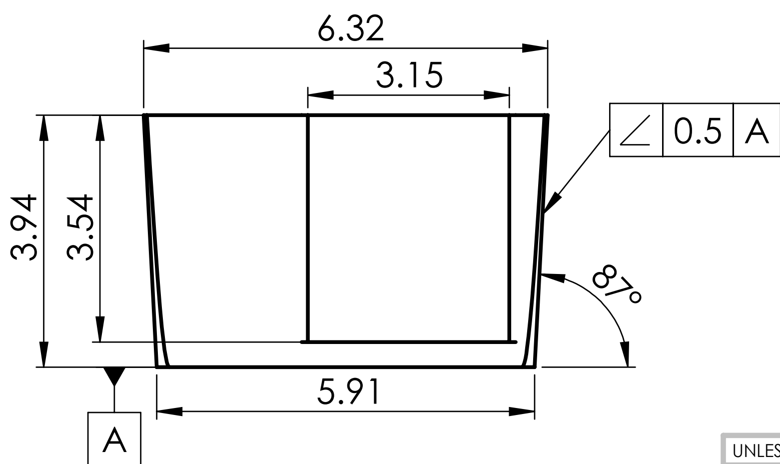


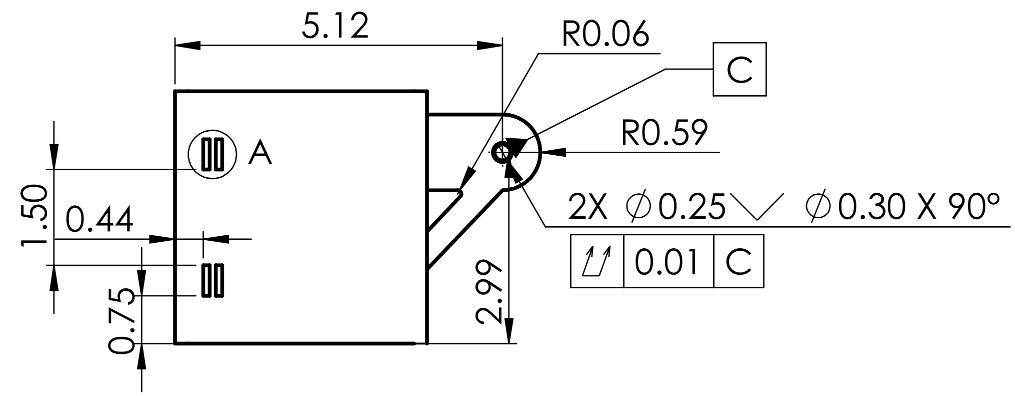
NOTES

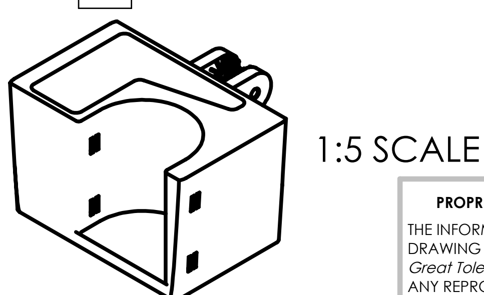
1. R = 0.020 UNLESS OTHERWISE STATED
2. ALL HOLES ARE THRU HOLES
3. ALL DETAIL A DIMENSIONS REPEATED FOR BOTTOM SLITS
4. ALL SLITS ARE THRU SLITS



DETAIL A
SCALE 2 : 3







1:5 SCALE

PICTORIAL FOR CLARITY

PROPRIETARY AND CONFIDENTIAL

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TOLERANCES:
FRACTIONAL ±1/32
TWO PLACE DECIMAL ±0.05
THREE PLACE DECIMAL ±0.015

INTERPRET GEOMETRIC TOLERANCING PER:
ASME Y14.5M-1994

MATERIAL
Nylon

FINISH
Acetone Smoothed

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DM	11/30
CHECKED	JW	12/02
ENG APPR.	YL	12/8
MFG APPR.		
Q.A.		

COMMENTS:
THIRD ANGLE PROJECTION

Great Tolerance Factory Organization
E29-G1028

TITLE:

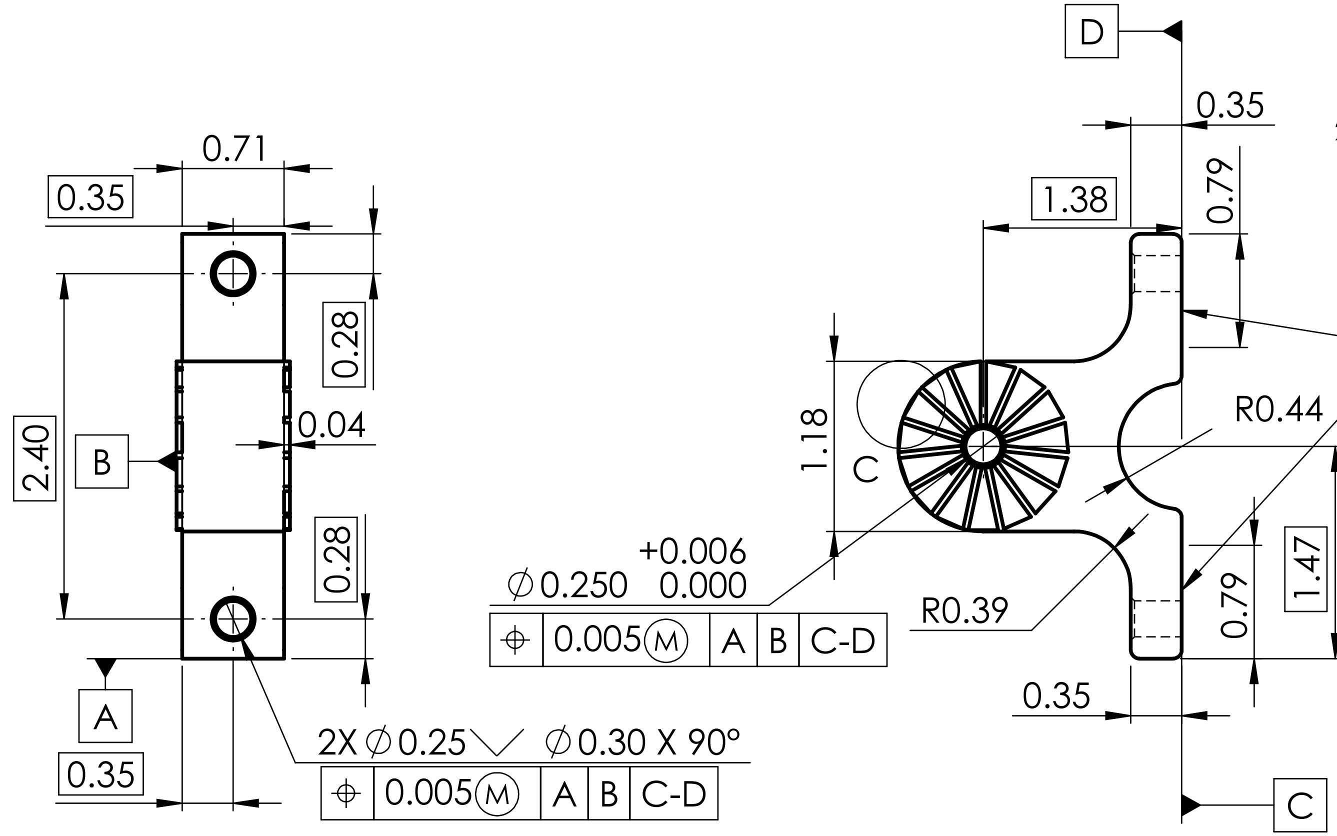
Main Tray

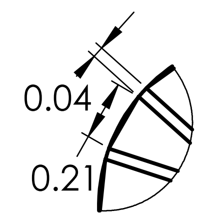
SIZE	PART NO.	REV
A	E29-G1028-1	3

SCALE: 1:3 SHEET 1 OF 1

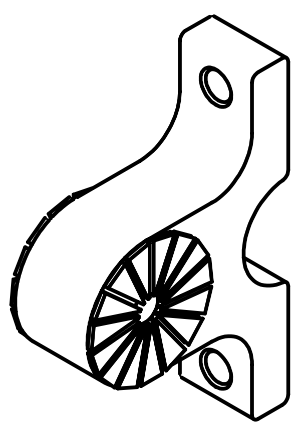
NOTES

1. ALL HOLES ARE THRU HOLES
2. R0.06 UNLESS STATED OTHERWISE





DETAIL C
SCALE 6 : 4



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DIMENSIONS ARE IN INCHES
TOLERANCES:

ALL DIMENSIONS: ±0.05

INTERPRET GEOMETRIC TOLERANCING PER:
ASME Y14.5M-1994

MATERIAL
Nylon

FINISH
Acetone Smoothed

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DM	11/30
CHECKED	JW	12/02
ENG APPR.	YL	12/8
MFG APPR.		
Q.A.		

COMMENTS:
THIRD ANGLE PROJECTION

Great Tolerance Factory Organization
E29-G1028

TITLE:

Clamp 1

SIZE **A** PART NO. E29-G1028-2 REV 3

SCALE: 3:4 SHEET 1 OF 1

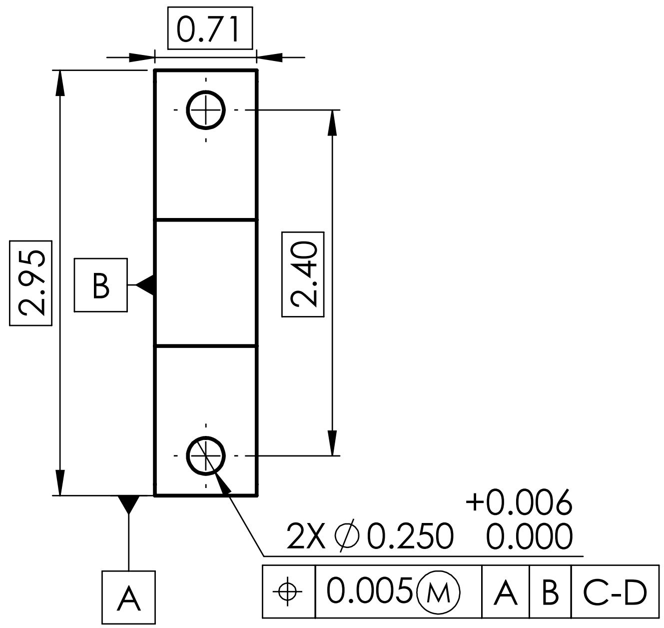
2 1

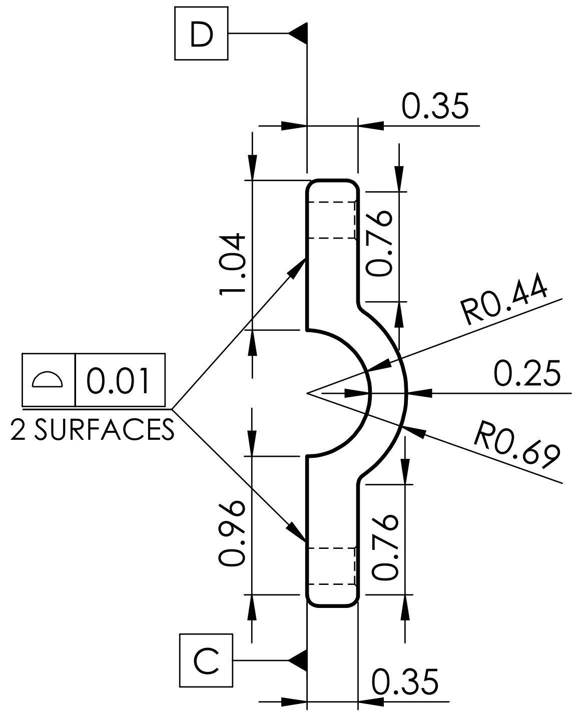
B B

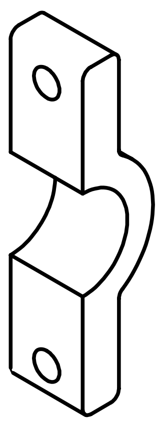
A A

NOTES

1. R0.08 if not stated
2. All holes are thru holes







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THREE PLACE DECIMAL ±0.015

INTERPRET GEOMETRIC TOLERANCING PER:
ASME Y14.5M-1994

MATERIAL
Nylon

FINISH
Acetone Smoothed

DO NOT SCALE DRAWING

	NAME	DATE
DRAWN	DM	11/30
CHECKED	JW	12/02
ENG APPR.	YL	12/8
MFG APPR.		
Q.A.		

COMMENTS:
THIRD ANGLE PROJECTION

Great Tolerance Factory Organization
E29-G1028

TITLE:

Clamp 2

SIZE	PART NO.	REV
A	E29-G1028-3	3

SCALE: 3:4 SHEET 1 OF 1

2 1

D.

Market Need

As students, an everyday challenge can be associated with transporting goods around campus, around town, and when commuting. Our group aims to offer a solution that could enrichen everyday life for students. The problem that our group decided to address is storage on human transportation.

Description of Project

Our product aims to solve the problem of storage on human transportation by creating a way for people to store their cups and small personal objects on a bike or scooter. Our product attaches to the handlebar of a scooter or bike, and adds significant space on it. It is designed to be compact, lightweight, and sturdy.

Differentiation

People currently solve our problem in very similar ways, using hooks or some sort of cup holder. An issue with other competitor's products is, it sacrifices practicality due to how uncomfortably bulky they are made. Our device is versatile due to its size and the fact that the angle of the storage tray can be adjusted with respect to the clamp. It can be used on either a bike or a scooter.

Fits and Tolerances table

Fit #	Connects Component # …(A)	…to Component # Or external object (B)	Function of fit (e.g. rotation joint)	ANSI fit class (e.g. RC6) or "reversible snap fit" or "permanent snap fit" with description of relative forces	Component A critical dimension (CD) and tolerance (tolerance in thousandths of an inch) e.g. diameter 10.0 ± 0.001 inches	Component B critical dimension and tolerance
I)	Clamp 1 (2)	HBOLT 0.2500-20 x2x0.75-N (8)	The bolt needs to pass through the hole in the clamp to reach the bolt on the other side.	Locational Clearance (LC10) fit: We want the bolt to be able to go through the hole in the clamp without interference and we do not need very tight tolerances on the hole.	CD: 0.25 (diameter) +6 0	CD: 0.25 (diameter) -5 -11
II)	Clamp 1 (2)	HBOLT 0.2500-20 x1x0.75-N (5)	The bolt needs to pass through the hole in the clamp to reach the bolt on the other side.	Locational Clearance (LC10) fit: We want the bolt to be able to go through the hole in the clamp without interference and we do not need very tight tolerances on the hole.	CD: 0.25 (diameter) +6 0	CD: 0.25 (diameter) -5 -11
III)	Clamp 2 (3)	HBOLT 0.2500-20 x1x0.75-N (5)	The bolt needs to pass through the hole in the clamp to reach the	Locational Clearance (LC10) fit: We want the bolt to be able to go through the hole in the clamp without	CD: 0.25 (diameter) +6 0	CD: 0.25 (diameter) -5 -11

			bolt on the other side.	interference and we do not need very tight tolerances on the hole.		
IV)	Main Tray (1)	HBOLT 0.2500-20 x2x0.75-N (8)	Attaches the tray to the clamp using a screw	Locational Clearance (LC10) fit: The tray is stationary after fixation and should fit snug without excessive play. The component should be assembled or disassembled freely.	CD: 0.25 (diameter) +6 0	CD: 0.25 (diameter) -5 -11
V)	18-8 Wing Nut (7)	HBOLT 0.2500-20 x2x0.75-N (8)	Connection between nut and bolt	Interference (LN1) fit: We want an interference fit between the nut and the bolt that goes through the tray mount hole.	CD: 0.25 (diameter) +0.4 0	CD: 0.25 (diameter) +0.65 +0.4
VI)	HNUT 0.2500-20-D-N (6)	HBOLT 0.2500-20 x1x0.75-N (5)	Connection between nut and bolt	Interference (LN1) fit: We want an interference fit between the nut and the bolt that goes through each clamp.	CD: 0.25 (diameter) +0.4 0	CD: 0.25 (diameter) +0.65 +0.4

Fits and Tolerances Table(s)

We decided to eliminate the cap to reduce complexity. For the fit between the holes on the clamps and the bolts, we decided to use a locational clearance fit (LC10). This fit allows for proper function of our parts while minimizing cost.

Additional Fits and Tolerances

For GD&T choices, we decided to first apply a position tolerance on the holes on the clamps to ensure the positions of the holes properly align with each other during the clamp assembly. Then, we chose to use a surface tolerance to make sure the outer surface of the clamps are flat enough to assemble. In addition, we indicate an angularity tolerance on the main tray to specify the draft angles during mass manufacturing processes, which is injection molding. Finally, a total runout is added to one of the holes on the main tray related to the cylindrical datum defined from the other hole. This ensures that the two holes are concentric and aligned within allowable tolerance.

Prototyping process selection

We decided to use fused deposition modeling to create our prototype. The reason we chose this manufacturing process is because our group has some 3D printing experience so it would be the most efficient way to make a prototype. Additionally, if we make any changes we can print our new prototype relatively quickly. For our prototype we decided to embed nuts into one of the clamps, but this is NOT reflected in the drawings because the drawings are catered to our design for mass production.

Scaled-up production plan

For full-scale production we would use injection molding due to the fact that it is proven, efficient, and cheap. This is the most appropriate mass manufacturing process for our product because each part is made using plastic and we do not have minimal undercuts that would not complicate the injection molding process too much. Once again, our drawings reflect our mass manufacturing design.

Material choices

For the main tray, we chose to use nylon as it is low per unit cost, can be manufactured using injection molding, and has material properties that make sense for our product. Specifically, it is strong yet has some flexibility. For the left clamp and right clamp, we chose to use nylon as well. The rubber band is an off-the- shelf item that we will be purchasing for our product. For the tray nut, tray bolt, clamp nut, and clamp bolt, these are also off-the-shelf items that we would purchase for our product. We would use carbon steel nuts and bolts due to lower energy cost and decent consistency thanks to use of forming die.

Design for Manufacturing

When designing our part, we came to the conclusion that we would use injection molding for mass manufacturing early on. With this in mind, we set our tolerances appropriately on our parts and we have a draft angle on the main tray.

Reflection

The biggest takeaway from this project is that designing, creating, and properly documenting a prototype is a difficult yet rewarding process. Communication between team members and accountability are both critical to accomplishing the task. Prototyping and testing our design showed us the importance of iteration and reflection in the process of making something new.

Just because something looks like it will work on paper does not always mean it will instantly work in real life. Unfortunately, our prototype broke before lab 14 began so we were not able to watch another group put it together. However, assembling the prototype that group 1025 created and seeing their engineering drawings allowed us to see things from the perspective of someone who did not create the prototype. This helped us to create our drawings for someone who is not familiar with our prototype. The most challenging task that we faced was the engineering drawings. Applying GD&T requires deeper understanding than answering test questions about it, and we struggled with it. One thing that stood out about the project is the importance of documentation. Any parts of our project that were not documented well caused trouble down the road if we needed to look back on what we did. Another aspect of this project that stood out is working as a team. Our team was not as cohesive as it could have been and sometimes team members did not contribute as planned, and others would have to step up. However, we were still able to accomplish everything in the end. In the future, we would establish the roles in the team early on and hold each other accountable. This project was a great learning experience in all of the aforementioned areas.

Course Evaluation